TEMAN BICARA - SYSTEM DESIGN & ERD

Backend Service Architecture, Role Interoperability & Database Schema

PLATFORM	DATABASE & AUTH	PAYMENT GATEWAY	COMMUNICATION
Next.js Fullstack (App Router)	Supabase (PostgreSQL + RLS)	Midtrans (Snap + Webhook)	Resend + Realtime Channel

1. Ikhtisar Ekosistem & Matriks 4 Role

Platform **Teman Bicara** menghubungkan pencari bantuan mental dengan profesional dan relawan pendengar melalui pemisahan peran yang tegas dan terisolasi.

ROLE 1: PASIEN (CLIENT)

- Login via Supabase Auth / Google OAuth.
- Curhat Anonim Gratis: Masuk antrean peer support bersama Relawan.
- Konseling Berbayar: Cari Psikolog, pilih jadwal, booking sesi.
- Pembayaran Midtrans Snap (QRIS, VA, E-Wallet, Kartu).
- Chat/Room Realtime & Rating/Review pasca-sesi.

ROLE 2: PSIKOLOG (PROFESIONAL)

- Onboarding & verifikasi lisensi (No. STR / SIP).
- Kelola ketersediaan jadwal slot konsultasi (harian/mingguan).
- Akses sesi konsultasi aktif (Chat/Video) dengan pasien.
- Isi Rekam Medis (SOAP Notes) terlindungi enkripsi & RLS.
- Dasbor saldo fee & pengajuan payout (penarikan dana).

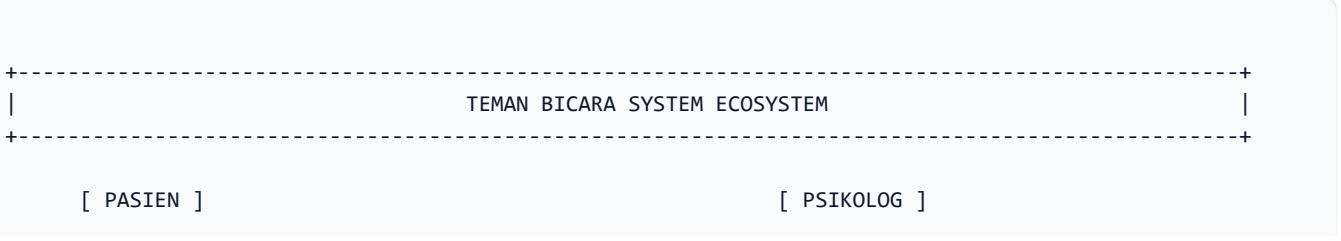
ROLE 3: RELAWAN (PEER LISTENER)

- Pendaftaran relawan pendengar + screening/pelatihan.
- Status toggle online/offline (siap mendengarkan).
- Klaim tiket curhat gratis dari antrean (Queue matching).
- Chat santai empatik anonim (tanpa rekam medis klinis).
- **Tombol Protokol Krisis:** Eskalasi indikasi self-harm/suicide ke admin & hotline darurat.

ROLE 4: ADMIN (SUPERVISOR)

- Verifikasi kelayakan STR/SIP Psikolog & screening Relawan.
- Monitoring audit transaksi Midtrans & approval payout.
- Monitoring eskalasi krisis darurat dari relawan/pasien.
- Manajemen suspensi pengguna & audit logging sistem.

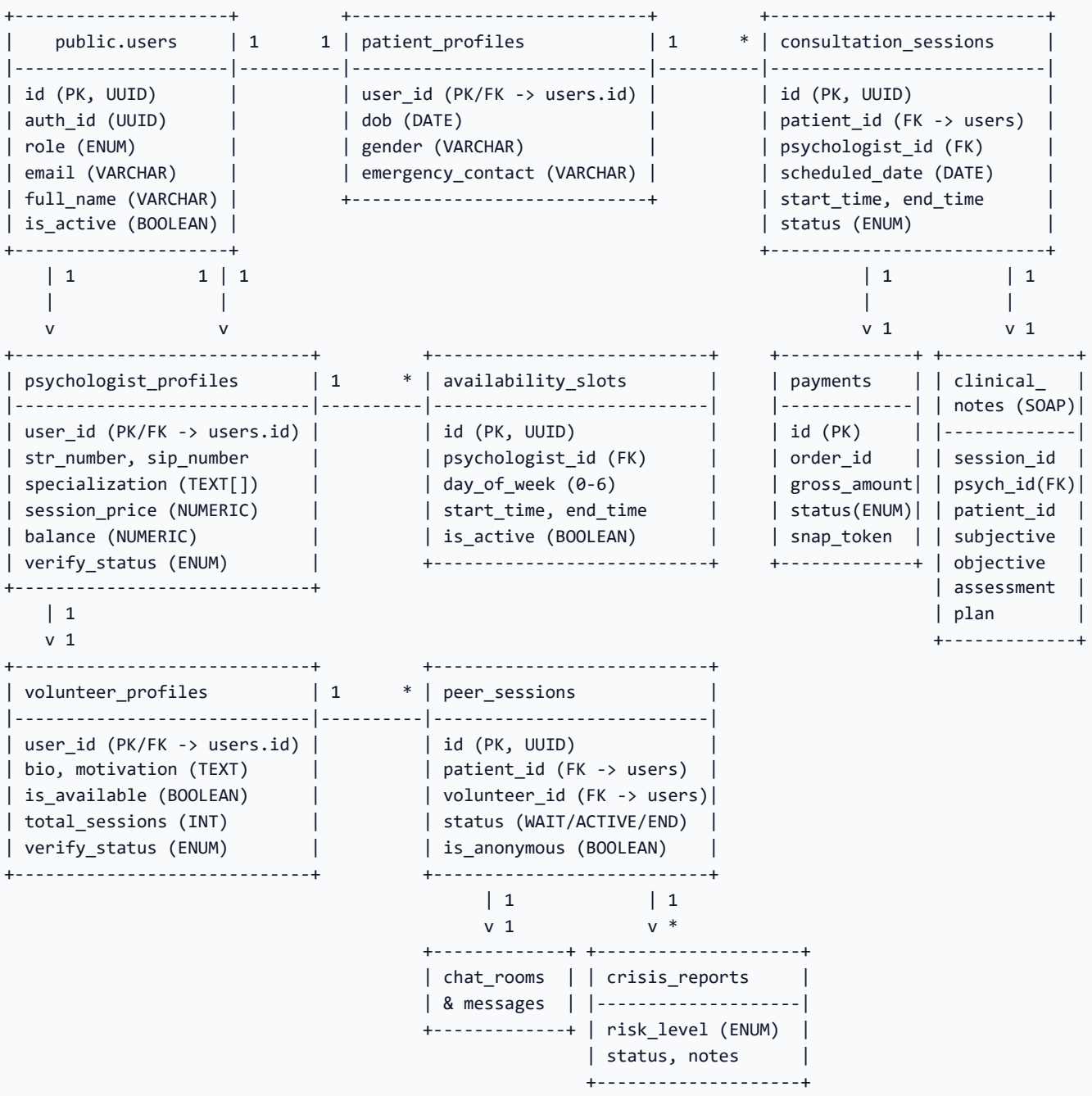
2. Use Case Diagram (Matriks Interaksi Antar-Role)




(1. Login / OAuth Google)
(2. Antrean Curhat Relawan)
[ROOM PEER CHAT] <==== (Sesi Curhat) ====> [RELAWAN]
[Trigger Krisis] ---> [ALERT KRISIS]
(3. Booking Konsul Psikolog)
[MIDTRANS SNAP]
(Webhook IPN)
[ROOM KONSUL PRIVAT]
(4. Beri Rating & Review)
(1. Registrasi STR/SIP)
(2. Set Jadwal Slot)
(3. Terima Sesi Konsul)
[SOAP CLINICAL NOTES]
(Strict RLS)
[SALDO & PAYOUT]
[APPROVE PAYOUT]
[ADMIN / SUPERVISOR]
(Verifikasi STR)
(Monitor Transaksi

3. Entity Relationship Diagram (ERD) & Kamus Data

Arsitektur data dirancang dengan pemisahan tabel profil untuk menjaga integritas relasi dan isolasi hak akses (RLS) PostgreSQL Supabase.



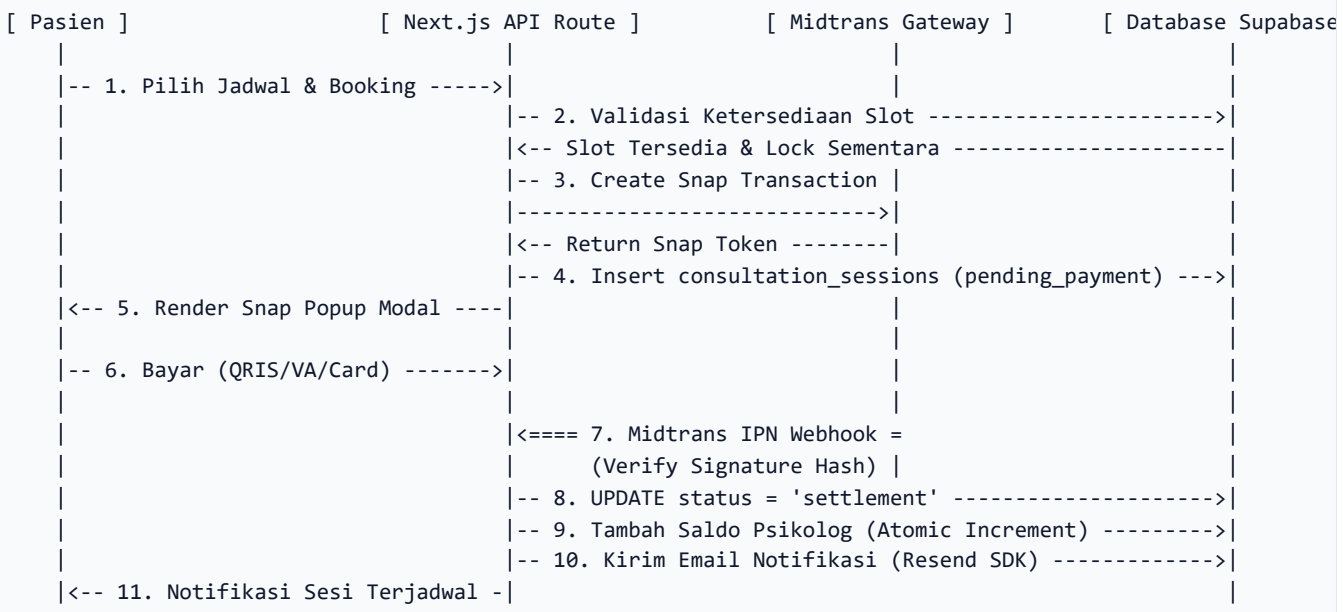
Daftar Entitas Utama & Struktur Field

Entitas (Tabel)	Field Kunci & Tipe Data	Aturan Integritas & RLS Security
`users`	`id (UUID PK), auth_id (UUID), email, role ('patient'|'psychologist'|'volunteer'|'admin'), full_name, is_active`	Sync otomatis via Supabase Trigger saat insert di `auth.users`. RLS: Read public profile, Update self.

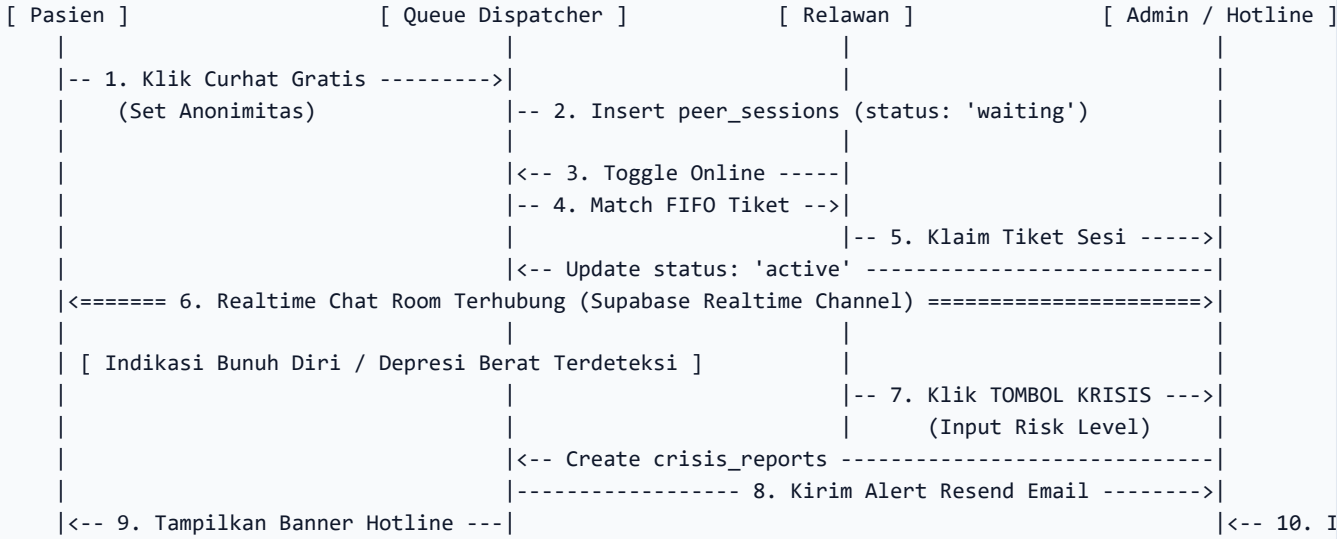
Entitas (Tabel)	Field Kunci & Tipe Data	Aturan Integritas & RLS Security
`psychologist_profiles`	`user_id (PK/FK), str_number, sip_number, specialization[], session_price, balance, verify_status`	Isolasi balance dan nomor STR. Hanya admin yang berhak mengubah `verify_status` ke 'approved'.
`volunteer_profiles`	`user_id (PK/FK), bio, is_available, verify_status, total_sessions`	Status availability hanya bisa diubah bila `verify_status = 'approved'`.
`consultation_sessions`	`id (UUID), patient_id (FK), psychologist_id (FK), scheduled_date, start_time, end_time, status`	Status: `pending_payment`, `scheduled`, `in_progress`, `completed`, `cancelled`.
`payments`	`id (UUID), session_id (FK), order_id (UNIQUE), gross_amount, transaction_status, snap_token, paid_at`	Idempotent webhook. Validasi signature SHA512 Midtrans sebelum update status menjadi `settlement`.
`clinical_notes`	`id (UUID), session_id (FK), psychologist_id (FK), patient_id (FK), subjective, objective, assessment, plan`	**STRICT RLS:** Pasien DILARANG membaca tabel ini. Hanya Psikolog pemilik sesi & Superadmin audit.
`peer_sessions`	`id (UUID), patient_id (FK), volunteer_id (FK), status ('waiting'|'active'|'ended'|'escalated'), is_anonymous`	Menangani antrean curhat gratis. Matching algoritma FIFO saat volunteer online.
`crisis_reports`	`id (UUID), session_id (FK), reported_by (FK), patient_id (FK), risk_level ('low'|'medium'|'high'|'severe'), status`	Trigger otomatis notifikasi email via Resend ke admin & hotline tim tanggap darurat saat status krisis dibuat.
`chat_messages`	`id (UUID), room_id (FK), sender_id (FK), content, attachment_url, is_read, created_at`	Di-stream langsung menggunakan Supabase Realtime Channel atau WebSocket. RLS: Hanya member room yang sah.

4. Flowchart Detail Sistem Teman Bicara

A. Flow 1: Booking Konseling Psikolog & Pembayaran Midtrans

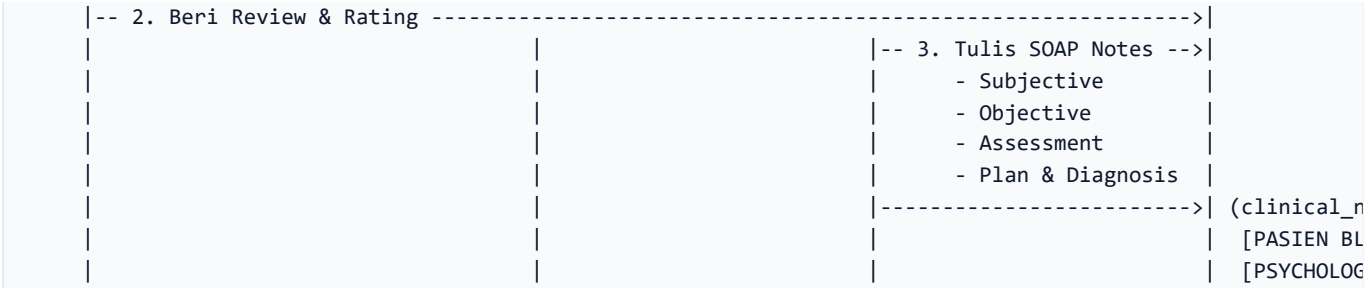


B. Flow 2: Antrean Curhat Relawan (Peer Support) & Protokol Krisis



C. Flow 3: Siklus Pasca-Konseling Psikolog & Rekam Medis (SOAP)




|-- 2. Beri Review & Rating ----------->|
|-- 3. Tulis SOAP Notes -->|
- Subjective
- Objective
- Assessment
- Plan & Diagnosis
|------------------------->| (clinical_n
[PASIEN BL
[PSYCHOLOG

5. Arsitektur Teknis & Roadmap Bertahap (Next.js + Supabase)

A. Manajemen Koneksi Realtime: Supabase Realtime vs WebSocket Standalone

> **Rekomendasi Arsitektur Lead Backend:**
> Karena backend menggunakan Next.js App Router (Serverless / Node Runtime) yang didukung Supabase, gunakan **Supabase Realtime Broadcast & Postgres Changes** secara native. Ini mengeliminasi kebutuhan mengelola server daemon `socket.io` terpisah, menghemat biaya infra, dan otomatis mewarisi autentikasi token JWT Supabase.

B. Desain Kebijakan Keamanan Row Level Security (RLS)

```
-- Proteksi Catatan Rekam Medis (SOAP): Pasien TIDAK BISA SELECT
ALTER TABLE clinical_notes ENABLE ROW LEVEL SECURITY;

CREATE POLICY "Psikolog hanya bisa kelola catatan buatannya"
ON clinical_notes FOR ALL
USING (auth.uid() = psychologist_id);

CREATE POLICY "Admin audit rekam medis"
ON clinical_notes FOR SELECT
USING (EXISTS (SELECT 1 FROM users WHERE users.id = auth.uid() AND users.role = 'admin'));

-- Pasien tidak punya hak SELECT ke clinical_notes
```

C. Roadmap Pengerjaan Bertahap (Phased Milestones)

Tahapan	Deliverable Backend	Detail Endpoint / Pekerjaan
Fase 1 (Fondasi)	Skema Basis Data & Autentikasi 4 Role	- DDL Migrasi Supabase (PostgreSQL schema, triggers, RLS). - Supabase Auth + Google OAuth handler. - Endpoint profil: `/api/auth/me`, `/api/profile/update`.
Fase 2 (Peer Support)	Fitur Curhat Relawan & Antrean Gratis	- Dispatcher antrean matching FIFO. - Supabase Realtime Channel untuk chat anonim. - Protokol tombol krisis: `/api/crisis/report` + Resend alert.
Fase 3 (Konseling Pro)	Jadwal Psikolog & Midtrans Payment	- CRUD Slot jadwal: `/api/psychologist/slots`. - Booking konsultasi & Snap token: `/api/booking/checkout`. - Webhook handler IPN Midtrans ber-signature: `/api/webhooks/midtrans`.
Fase 4 (Klinis & Payout)	SOAP Clinical Notes & Finansial	- API Catatan Medis (SOAP) ter-enkripsi. - Perhitungan saldo komisi platform. - Pengajuan & verifikasi payout: `/api/payouts/request`.
Fase 5 (Backoffice)	Admin Portal & Audit	- Approval lisensi STR/SIP Psikolog. - Audit transaksi & laporan pendapatan platform.

Perhatian Kredensial Produksi:
Key dan token rahasia (`SUPABASE_SERVICE_ROLE_KEY`, `PAYMENT_SERVER_KEY`, `RESEND_API_KEY`, `JWT_SECRET`) hanya boleh dipanggil di Server-side Route Handler (`app/api/*`). Dilarang membocorkan variabel tanpa prefix `NEXT_PUBLIC_` ke sisi client bundle.